

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
 Department of Computer Science and Engineering
ASSESSMENT TOOL 2 – SCENARIO BASED ASSESSMENT (2 Questions)

Course Code:	ITA05	Course Title:	COMPUTER VISION
CO Assessed:	CO2 – Manipulation (BL3)	Assessment:	Assessment Tool 2 – Scenario Based Assessment
Weightage:	20% of CIA	Total Marks:	20 (2 Questions × 10 Marks)
CO5 Statement:	Apply image enhancement and segmentation techniques for extracting meaningful regions from images.(BL3)		
Student Name:	Navya.Y	Reg. No.:	192425177
Section / Batch:	Slot-c	Date:	

Code of Conduct

I _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: _____

Instructions

- This test contains 2 questions, each carrying 10 marks. Total: 20 Marks.
- No negative marking. Unanswered questions carry zero marks.
- No calculators or electronic devices permitted.
- Duration: 60 minutes.

Section / Topic	Focus Area	Q. Nos.	Marks
Frequency Domain Image Enhancement	Periodic noise removal, Fourier Transform, notch filtering, quality inspection	1	10
Spatial Filtering & Edge-Preserving Enhancement	Bilateral filtering, edge preservation, autonomous vehicle navigation	2	10
GRAND TOTAL:			20 Marks

Annexure A – Scenario Based Assessment

Scenario 1: Manufacturing Quality Inspection

An automated quality inspection system captures images of metal components. Electrical interference introduces repetitive horizontal stripes across the images, making defect detection unreliable.

- (a) Interpret the scenario and identify the type of noise present.
- (b) Explain how this degradation affects inspection accuracy.
- (c) Recommend an appropriate enhancement technique to remove the interference.
- (d) Justify why your selected approach is more suitable than spatial-domain methods.
- (e) Explain your answer in a logical sequence.

Scenario 2: Autonomous Vehicle Navigation

A self-driving car captures road images during heavy rain. Although noise is present, the lane markings and road edges must remain sharp for safe navigation.

- (a) Interpret the scenario and explain the challenge involved.
- (b) Identify the importance of preserving edges while removing noise.
- (c) Recommend a suitable filtering technique.
- (d) Justify why your selected filter preserves important image details.
- (e) Present your explanation clearly and systematically.

Rubrics for Calculation

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)	Marks
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Understanding of Scenario	20%	Demonstrates clear and complete understanding of the scenario and context	Shows good understanding with minor gaps	Partial understanding ; misses key aspects	Misinterprets or fails to understand the scenario		
Identification of Key Issues	20%	Accurately identifies all relevant issues and factors involved	Identifies most key issues with minor omissions	Identifies limited issues; some irrelevant points	Unable to identify key issues		
Application of Concepts	25%	Applies appropriate concepts effectively to address the scenario	Applies concepts with minor errors	Limited or partially correct application	Unable to apply relevant concepts		
Decision Making	20%	Makes well-reasoned, logical decisions with strong rationale	Makes reasonable decisions with some justification	Makes weak decisions with limited justification	No clear decisions made or rationale provided		
Clarity of Response	15%	Response is clear, structured, and logically presented	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand		

Faculty In-charge	Course Coordinator	Head of Department
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

Scenario 1: Manufacturing Quality Inspection

(a) Identify the type of noise present

The repetitive horizontal stripes caused by electrical interference indicate Periodic Noise. This type of noise appears as regular horizontal or vertical patterns due to electrical or electronic interference during image acquisition.

(b) Effect on inspection accuracy

- Hides or distorts small defects such as scratches and cracks.
- Reduces image clarity and contrast.
- Causes false detection or missed defects.
- Lowers the accuracy of automated quality inspection systems.

(c) Appropriate enhancement technique

Use a Frequency Domain Notch Filter (Band-Reject/Notch Reject Filter).

(d) Justification

- Periodic noise appears as distinct peaks in the Fourier Transform (frequency domain).
- A notch filter removes only the unwanted frequency components while preserving the actual image details.
- Spatial-domain filters (mean, median, Gaussian) cannot effectively remove periodic stripe patterns and often blur important features and edges.

Therefore, a frequency-domain notch filter is more suitable than spatial-domain methods.

(e) Logical sequence

1. The inspection system captures images of metal components.
2. Electrical interference introduces repetitive horizontal stripe patterns.
3. The stripes are identified as periodic noise.
4. The noise obscures defects and reduces inspection accuracy.

5. Apply the Fourier Transform to convert the image into the frequency domain.
6. Detect the noise frequencies and remove them using a notch filter.
7. Apply the inverse Fourier Transform to reconstruct a clean image.
8. The enhanced image improves defect visibility and increases the accuracy of automated quality inspection.

Scenario 2: Autonomous Vehicle Navigation

(a) Interpret the scenario and explain the challenge

A self-driving car captures road images during heavy rain, introducing noise into the images. The challenge is to remove the noise while preserving lane markings, road edges, and other important features, as these are essential for safe navigation.

(b) Importance of preserving edges while removing noise

- Lane markings help the vehicle stay in the correct lane.
- Road edges define the driving boundaries.
- Sharp edges improve object and obstacle detection.
- Preserving edges ensures accurate image analysis and safe autonomous driving.

(c) Suitable filtering technique

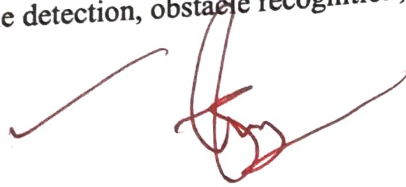
Use a Bilateral Filter.

(d) Justification

- The bilateral filter reduces noise while preserving edges.
- It considers both the spatial distance and pixel intensity differences when smoothing the image.
- Pixels across edges are not averaged together, so important boundaries remain sharp.
- Unlike mean or Gaussian filters, it does not significantly blur edges.

(e) Systematic explanation

1. The self-driving car captures road images during heavy rain.
2. Rain introduces noise into the captured images.
3. Lane markings and road edges must remain clear for navigation.
4. Apply a bilateral filter to reduce noise.
5. The filter smooths similar neighboring pixels while preserving edge boundaries.
6. The enhanced image retains sharp lane markings and road edges.
7. This improves lane detection, obstacle recognition, and overall driving safety.

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